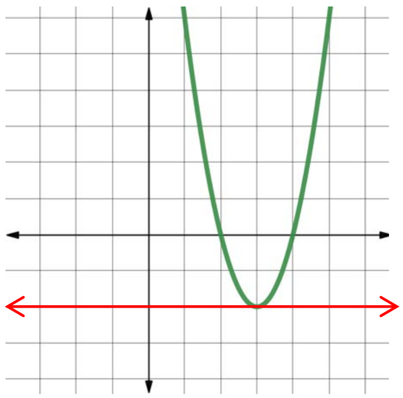
Algebra 1
Unit 9
Lesson 5 Practice

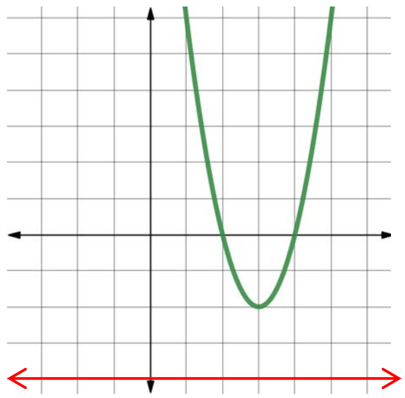
Name____Answer key_____
Date___________________
Period__________________

Draw a line on the graph of the quadratic where there would be…

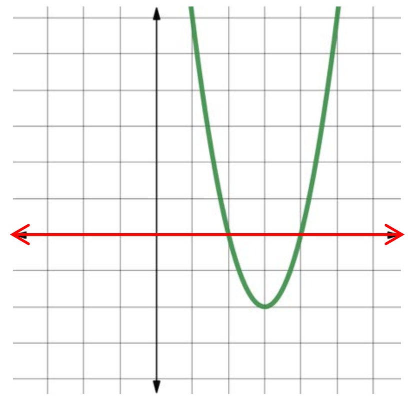
1. …one real solution.



2. …no real solution.



3. …two real solutions.



Solve each equation. Show your work.

4. $3x^2 = 48$

$\frac{3x^2}{3} = \frac{48}{3}$

$\sqrt{x^2} = \sqrt{16}$

$x = \pm 4$

5. $-20 = -2x^2$

$\frac{-20}{-2} = \frac{-2x^2}{-2}$

$\sqrt{10} = \sqrt{x^2}$

$x = \pm\sqrt{10}$

6. $\frac{1}{2}x^2 = 2$

$2\left(\frac{1}{2}x^2 = 2\right)$

$\sqrt{x^2} = \sqrt{4}$

$x = \pm 2$

7. $2(x^2+4)=80$

$\frac{2(x^2+4)}{2}=\frac{80}{2}$

$x^2+\cancel{4}=40$
$\underline{\cancel{-4}}\quad \underline{-4}$

$\sqrt{x^2}=\sqrt{36}$ $\qquad x=\pm 6$

8. $55=x^2+6$

$55=x^2+6$
$\underline{-6}\qquad \underline{-6}$
$49=x^2$
$\sqrt{49}=\sqrt{x^2}$ $\qquad x=\pm 7$

9. $2x^2+5=55$

$2x^2+5=55$
$\underline{-5}\quad \underline{-5}$
$\frac{2x^2+\emptyset}{2}=\frac{50}{2}$ $\qquad \sqrt{x^2}=\sqrt{25}$
$x=\pm 5$

10. $x^2-1=2$

$x^2-1=2$
$\underline{+1}\quad \underline{+1}$
$x^2=3$
$\sqrt{x^2}=\sqrt{3}$

$x=\pm\sqrt{3}$